



# ENVIRONMENTAL PRODUCT DECLARATION

IN ACCORDANCE WITH EN 15804+A2 & ISO 14025

Lo-Carbon Sentinel Kinetic  
Volution Ventilation



**EPD HUB, HUB-5701**

Published on 13.03.2026, last updated on 13.03.2026, valid until 12.03.2031

Life Cycle Assessment study has been performed in accordance with the requirements of EN 15804, EPD Hub PCR version 1.2 (24 Mar 2025) and JRC characterization factors EF 3.1.



Created with One Click LCA

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Healthy air, sustainably.

## GENERAL INFORMATION

### MANUFACTURER

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| Manufacturer    | Volution Ventilation                                       |
| Address         | Fleming Way, CRAWLEY, West Sussex, United Kingdom RH10 9YX |
| Contact details | EPD@Volution-group.co.uk                                   |
| Website         | www.volutiongroupplc.com                                   |

### EPD STANDARDS, SCOPE AND VERIFICATION

|                    |                                                                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program operator   | EPD Hub, hub@epdhub.com                                                                                                                                                                |
| Reference standard | EN 15804:2012+A2:2019/AC:2021 and ISO 14025                                                                                                                                            |
| PCR                | EPD Hub Core PCR Version 1.2, 24 Mar 2025                                                                                                                                              |
| Sector             | Manufactured product                                                                                                                                                                   |
| Category of EPD    | Third party verified EPD                                                                                                                                                               |
| Parent EPD number  | -                                                                                                                                                                                      |
| Scope of the EPD   | Cradle to gate with options, A4-A5, B6 and modules C1-C4, D                                                                                                                            |
| EPD author         | Andrew Jackson                                                                                                                                                                         |
| EPD verification   | Independent verification of this EPD and data, according to ISO 14025:<br><br><input type="checkbox"/> Internal verification <input checked="" type="checkbox"/> External verification |
| EPD verifier       | Yazan Badour as an authorized verifier for EPD Hub                                                                                                                                     |

This EPD is intended for business-to-business and/or business-to-consumer communication. The manufacturer has the sole ownership, liability, and responsibility for the EPD. EPDs within the same product category but from different programs may not be comparable. EPDs of construction products may not be comparable if they do not comply with EN 15804 and if they are not compared in a building context.

## PRODUCT

|                                            |                            |
|--------------------------------------------|----------------------------|
| Product name                               | Lo-Carbon Sentinel Kinetic |
| Additional labels                          | -                          |
| Product reference                          | See Annex 1                |
| Place(s) of raw material origin            | UK, China, Europe          |
| Place of production                        | Dudley, United Kingdom     |
| Place(s) of installation and use           | UK, Europe                 |
| Period for data                            | 01-09-2024 to 31-08-2025   |
| Averaging in EPD                           | No grouping                |
| Variation in GWP-fossil for A1-A3 (%)      | -                          |
| GTIN (Global Trade Item Number)            | See Annex 1                |
| NOBB (Norwegian Building Product Database) | -                          |
| A1-A3 Specific data (%)                    | 0.69                       |

## ENVIRONMENTAL DATA SUMMARY

|                                             |          |
|---------------------------------------------|----------|
| Declared unit                               | 1 Unit   |
| Declared unit mass                          | 25.34 kg |
| Mass of packaging                           | 16 kg    |
| GWP-fossil, A1-A3 (kgCO <sub>2</sub> e)     | 320      |
| GWP-total, A1-A3 (kgCO <sub>2</sub> e)      | 294      |
| Secondary material, inputs (%)              | 7.53     |
| Secondary material, outputs (%)             | 23.4     |
| Total energy use, A1-A3 (kWh)               | 1200     |
| Net freshwater use, A1-A3 (m <sup>3</sup> ) | 3.11     |

# PRODUCT AND MANUFACTURER

## ABOUT THE MANUFACTURER

Founded in 1936, Vent-Axia improves the indoor environment by providing ventilation solutions across the Residential and commercial New build, Repair and maintenance segments of the UK construction market with a focus in Private and Public housing, New Build developments and Commercial and Industrial buildings. In December 2002, Volution Group was formed through a buy-out. In June 2014 Volution Group PLC became a listed company on the London Stock Exchange (LSE: FAN). Volution have acquired many brands over the years in the UK, mainland Europe, Australia and New Zealand.

## PRODUCT DESCRIPTION

The Lo-Carbon Sentinel Kinetic models can be mounted vertically in a roof space or in an appropriate cupboard within the dwelling. When mounted in an unheated area the ducting and unit must be insulated in accordance with the Domestic Ventilation Compliance Guide. Ducting can be attached to the unit horizontally, vertically or both.

For scenarios where noise is a critical issue, an Acoustic Enclosure is available on certain models to reduce breakout noise and the Acoustic Top Box will reduce in-duct noise at key frequencies. Left (L) or right (R) hand installation. Left hand and right hand units are available on certain models.

The combination of spigot options allows installation in confined locations. The condensate drain can be taken out through the back, side or bottom of the unit. Using the fittings supplied, the final condensate connection is made outside the unit and can be completed after installation.

The integral humidity sensor increases speed in proportion to relative humidity levels, saving energy and reducing noise. The sensor also reacts to small but rapid increases in humidity, even if the normal trigger threshold is

not reached. This unique feature ensures adequate ventilation, even for the smallest wet room. The night time relative humidity setback feature suppresses nuisance tripping as humidity gradually increases with falling temperature.

Further information can be found at:  
[www.volutiongroupplc.com](http://www.volutiongroupplc.com)

## PRODUCT RAW MATERIAL MAIN COMPOSITION

| Raw material category | Amount, mass % | Material origin |
|-----------------------|----------------|-----------------|
| Metals                | 26             | EU, Asia & UK   |
| Minerals              | 24             | EU & UK         |
| Fossil materials      | 50             | EU & UK         |
| Bio-based materials   | -              | -               |

## BIOGENIC CARBON CONTENT

Product's biogenic carbon content at the factory gate

|                                            |      |
|--------------------------------------------|------|
| Biogenic carbon content in product, kg C   | 0    |
| Biogenic carbon content in packaging, kg C | 7.42 |

## FUNCTIONAL UNIT AND SERVICE LIFE

|                        |          |
|------------------------|----------|
| Declared unit          | 1 Unit   |
| Mass per declared unit | 25.34 kg |
| Functional unit        | -        |
| Reference service life | 20       |

## SUBSTANCES, REACH - VERY HIGH CONCERN

The product does not contain any REACH SVHC substances in amounts greater than 0,1 % (1000 ppm).

## PRODUCT LIFE-CYCLE

### SYSTEM BOUNDARY

This EPD covers the life-cycle modules listed in the following table.

| Product stage |           |               | Assembly stage |          | Use stage |             |        |             |               |                        |                       | End of life stage          |           |                  |          | Beyond the system boundaries |           |
|---------------|-----------|---------------|----------------|----------|-----------|-------------|--------|-------------|---------------|------------------------|-----------------------|----------------------------|-----------|------------------|----------|------------------------------|-----------|
| A1            | A2        | A3            | A4             | A5       | B1        | B2          | B3     | B4          | B5            | B6                     | B7                    | C1                         | C2        | C3               | C4       | D                            |           |
| x             | x         | x             | x              | x        | ND        | ND          | ND     | ND          | ND            | x                      | ND                    | x                          | x         | x                | x        | x                            |           |
| Raw materials | Transport | Manufacturing | Transport      | Assembly | Use       | Maintenance | Repair | Replacement | Refurbishment | Operational energy use | Operational water use | Deconstruction/ demolition | Transport | Waste processing | Disposal | Reuse                        | Recycling |

Not declared = ND.

### MANUFACTURING AND PACKAGING (A1-A3)

The environmental impacts considered for the product stage cover the manufacturing of raw materials used in the production as well as packaging materials and other ancillary materials. Also, fuels used by machines, and handling of waste formed in the production processes at the manufacturing facilities are included in this stage. The study also considers the material losses occurring during the manufacturing processes as well as losses during electricity transmission.



A market-based approach is used in modelling the electricity mix utilized in the factory.

The reference product consists of one indoor unit that is made of metals, plastics and electronic components. The materials are transported to our production facility, where the main manufacturing processes include sheet metal pressing and plastic moulding before being assembled. The manufacturing process requires electricity and fuels for the different equipment as well as heating. The manufacturing energy is considered based on the electricity mix of the manufacturing site, energy source and country of electricity generation. Certain ancillary materials are also included. Each part manufacturing process is considered separately and process waste disposal is accounted based on allocation of factory level waste. The finished product is packaged in cardboard before being sent to the installation site on a wooden pallet.

The manufacturing waste has been excluded as the quantity is less than 1% of the total material flow.

The use of green energy in manufacturing is demonstrated through contractual instruments (GOs, RECs, etc.), and its use is ensured throughout the validity period of this EPD.

### TRANSPORT AND INSTALLATION (A4-A5)

Transportation impacts occurred from final products delivery to construction site (A4) cover fuel direct exhaust emissions, environmental impacts of fuel production, as well as related infrastructure emissions.

A sales volume-based weighted average is considered for defining the distribution to the customer. Average distance of transportation from production plant to building site was calculated using our most common country of sale and using the capital city of this country as the destination. Where applicable, we have factored in a shipping leg for transportation into mainland Europe from the UK. Vehicle capacity utilisation volume factor is

assumed to be 100% which means full load. In reality, it may vary but as role of transportation emissions in total results is small, the variety in load is assumed to be negligible. To be conservative, empty returns are considered as modelled in the ecoinvent database. Transportation does not cause losses as product are packaged properly. Also, volume capacity utilisation factor is assumed to be 100% for the nested packaged products.

Transportation impacts that occur from delivery of the product cover direct exhaust emissions of fuel, environmental impacts of fuel production, as well as related infrastructure emissions. Environmental impacts from installation into the building include waste packaging materials (A5) and release of biogenic carbon dioxide from waste processing of cardboard and wood pallets. Electricity consumption, steel bolts and copper pipes are considered as part of the installation process. Quantities are assumed as it varies for different products.

The nuts, bolts and screws are part of the manufacturing process to hold the unit together. No screws are supplied to install the unit in a fixed location.

### PRODUCT USE AND MAINTENANCE (B1-B7)

The reference service life is 20 years and the total energy consumption during the use phase is also considered based on the sales volume-based weighted average data. No replacement of components or parts are included.

This EPD follows additional requirements for products using energy in B6 (Use Stage) permanently installed into the building.

Air, soil, and water impacts during the use phase have not been studied.

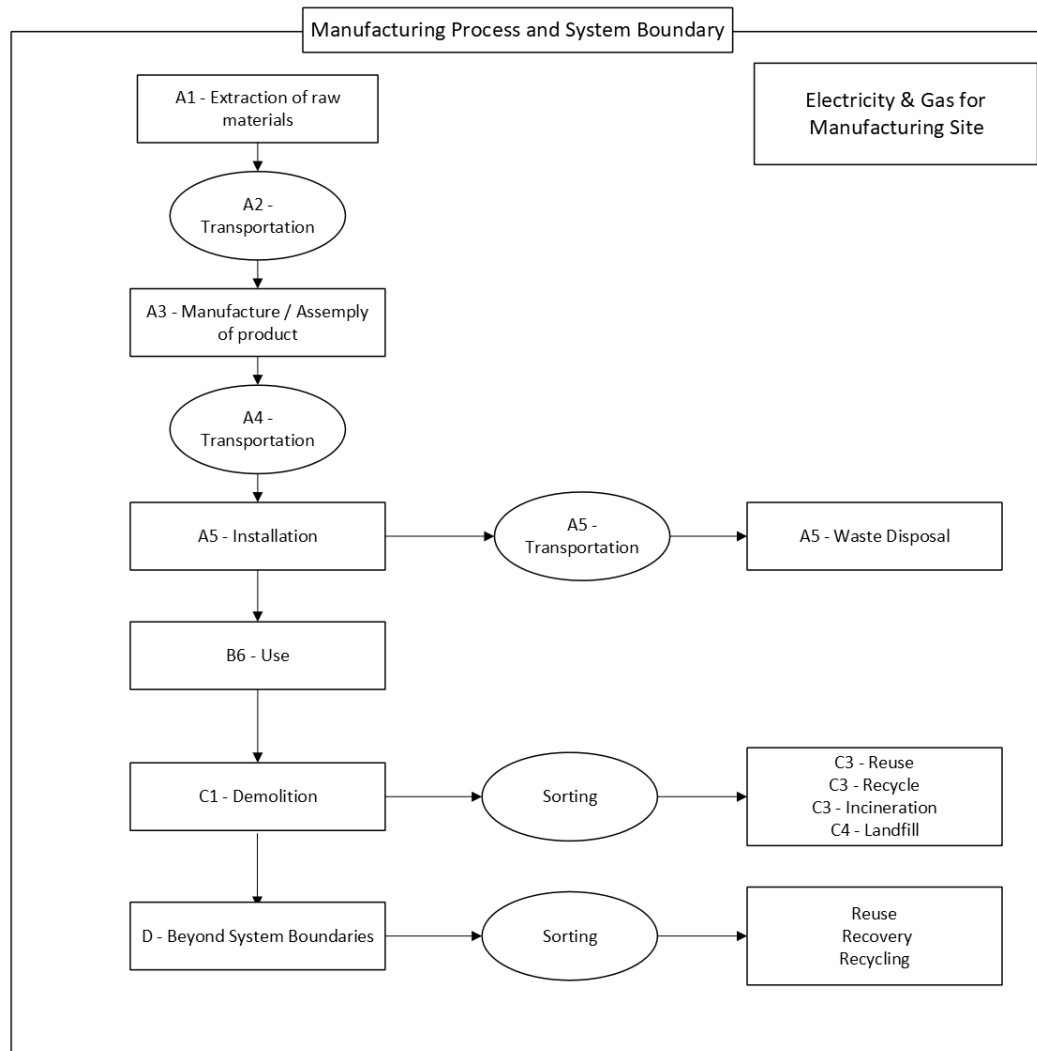
### PRODUCT END OF LIFE (C1-C4, D)

Consumption of energy and natural resources in demolition process is assumed to be negligible. It is assumed that the waste is collected

separately and transported to the waste treatment centre. Transportation distance to treatment is assumed as 250 km to recycling sites, 150 km to incineration and 50 km to landfill, and the transportation method is assumed to be lorry (C2). The used installation resources such as copper pipes and steel bolts are also dismantled and transported along with the product. According to EN 50693:2019, the sequence of treatment operations occurring to the product shall include de-pollution, fractions separation and preparation (dismantling, crushing, shredding, sorting), recycling, other material recovery, energy recovery and disposal. In this study, the default values from table G.4 of EN 50693 is used for treating materials in different waste treatment methods.

Benefits and loads beyond the system boundary were taken into consideration in this study. Specifically, the loads from recycling and incineration of the raw materials in the end of life, and the packaging in A5. Any recycled contents in the materials were discounted in Module D as per the EN 15804+A2.

## SYSTEM DIAGRAM





## LIFE-CYCLE ASSESSMENT

### CUT-OFF CRITERIA

The study does not exclude any modules or processes which are stated mandatory in the reference standard and the applied PCR. The study does not exclude any hazardous materials or substances. The study includes all major raw material and energy consumption. All inputs and outputs of the unit processes, for which data is available for, are included in the calculation. There is no neglected unit process more than 1% of total mass or energy flows. The module specific total neglected input and output flows also do not exceed 5% of energy usage or mass.

The production of capital equipment, construction activities, and infrastructure, maintenance and operation of capital equipment, personnel-related activities, energy and water use related to company management and sales activities are excluded.

### VALIDATION OF DATA

Data collection for production, transport, and packaging was conducted using time and site-specific information, as defined in the general information section on page 1 and 2. Upstream process calculations rely on generic data as defined in the Bibliography section. Manufacturer-provided specific and generic data were used for the product's manufacturing stage. The analysis was performed in One Click LCA EPD Generator, with the 'Cut-Off, EN 15804+A2' allocation method, and characterization factors according to EN 15804:2012+A2:2019/AC:2021 and JRC EF 3.1.

### ALLOCATION, ESTIMATES AND ASSUMPTIONS

Allocation is required if some material, energy, and waste data cannot be measured separately for the product under investigation. All allocations are done as per the reference standards and the applied PCR. In this study, allocation has been done in the following ways:

| Data type                      | Allocation                  |
|--------------------------------|-----------------------------|
| Raw materials                  | No allocation               |
| Packaging material             | Allocated by mass or volume |
| Ancillary materials            | Not applicable              |
| Manufacturing energy and waste | Allocated by mass or volume |

### PRODUCT & MANUFACTURING SITES GROUPING

|                                      |                |
|--------------------------------------|----------------|
| Type of grouping                     | No grouping    |
| Grouping method                      | Not applicable |
| Variation in GWP-fossil for A1-A3, % |                |

This EPD is product and factory specific.

## LCA SOFTWARE AND BIBLIOGRAPHY

This EPD has been created using One Click LCA EPD Generator for EPD Hub V3 and EPD Process Certification v3.2.4. The LCA and EPD have been prepared according to the reference standards and ISO 14040/14044. The EPD Generator uses Ecoinvent v3.10.1/3.11/3.12 and One Click LCA databases as sources of environmental data. Allocation used in Ecoinvent 3.10.1/3.11/3.12 environmental data sources follow the methodology 'allocation, Cut-off, EN 15804+A2'.

## ENVIRONMENTAL IMPACT DATA

The estimated impact results are only relative statements which do not indicate the end points of the impact categories, exceeding threshold values, safety margins or risks.

### CORE ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2, EF 3.1

| Impact category                     | Unit                   | A1       | A2       | A3        | A1-A3     | A4       | A5       | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3        | C4        | D         |
|-------------------------------------|------------------------|----------|----------|-----------|-----------|----------|----------|----|----|----|----|----|----------|----|----------|----------|-----------|-----------|-----------|
| GWP – total <sup>1)</sup>           | kg CO <sub>2</sub> e   | 3.11E+02 | 1.66E+00 | -1.80E+01 | 2.94E+02  | 1.68E+00 | 2.75E+01 | ND | ND | ND | ND | ND | 6.25E-06 | ND | 0.00E+00 | 6.31E-01 | 2.25E+01  | 1.14E+01  | -2.47E+01 |
| GWP – fossil                        | kg CO <sub>2</sub> e   | 3.09E+02 | 1.66E+00 | 8.70E+00  | 3.20E+02  | 1.68E+00 | 5.58E-01 | ND | ND | ND | ND | ND | 6.24E-06 | ND | 0.00E+00 | 6.31E-01 | 2.25E+01  | 1.14E+01  | -2.47E+01 |
| GWP – biogenic                      | kg CO <sub>2</sub> e   | 5.72E-01 | 2.75E-04 | -2.70E+01 | -2.64E+01 | 2.78E-04 | 2.70E+01 | ND | ND | ND | ND | ND | 3.65E-09 | ND | 0.00E+00 | 1.38E-04 | -3.66E-02 | -9.21E-04 | -2.17E-02 |
| GWP – LULUC                         | kg CO <sub>2</sub> e   | 8.51E-01 | 7.06E-04 | 2.65E-01  | 1.12E+00  | 7.63E-04 | 6.38E-04 | ND | ND | ND | ND | ND | 7.77E-09 | ND | 0.00E+00 | 2.80E-04 | 5.98E-04  | 1.39E-04  | 1.76E-02  |
| Ozone depletion pot.                | kg CFC <sub>11</sub> e | 9.88E-06 | 2.70E-08 | 2.42E-07  | 1.02E-05  | 2.47E-08 | 7.82E-09 | ND | ND | ND | ND | ND | 3.16E-13 | ND | 0.00E+00 | 8.86E-09 | 1.61E-08  | 8.61E-09  | -6.25E-08 |
| Acidification potential             | mol H <sup>+</sup> e   | 1.79E+00 | 6.34E-03 | 4.19E-02  | 1.84E+00  | 1.28E-02 | 2.77E-03 | ND | ND | ND | ND | ND | 2.32E-08 | ND | 0.00E+00 | 2.11E-03 | 1.38E-02  | 6.44E-03  | -1.36E-01 |
| EP-freshwater <sup>2)</sup>         | kg Pe                  | 2.24E-01 | 1.22E-04 | 4.25E-03  | 2.28E-01  | 1.19E-04 | 1.39E-04 | ND | ND | ND | ND | ND | 1.12E-09 | ND | 0.00E+00 | 4.91E-05 | 1.85E-04  | 5.70E-05  | -1.23E-02 |
| EP-marine                           | kg Ne                  | 3.13E-01 | 1.88E-03 | 1.87E-02  | 3.33E-01  | 3.60E-03 | 3.44E-03 | ND | ND | ND | ND | ND | 5.02E-09 | ND | 0.00E+00 | 6.83E-04 | 9.63E-03  | 9.98E-03  | -2.69E-02 |
| EP-terrestrial                      | mol Ne                 | 3.30E+00 | 2.05E-02 | 1.44E-01  | 3.46E+00  | 3.96E-02 | 1.05E-02 | ND | ND | ND | ND | ND | 5.81E-08 | ND | 0.00E+00 | 7.44E-03 | 6.80E-02  | 3.34E-02  | -2.75E-01 |
| POCP (“smog”) <sup>3)</sup>         | kg NMVOCe              | 1.11E+00 | 8.46E-03 | 4.59E-02  | 1.17E+00  | 1.31E-02 | 3.59E-03 | ND | ND | ND | ND | ND | 1.62E-08 | ND | 0.00E+00 | 2.95E-03 | 1.70E-02  | 8.43E-03  | -8.93E-02 |
| ADP-minerals & metals <sup>4)</sup> | kg Sbe                 | 3.74E-02 | 4.81E-06 | 3.92E-05  | 3.75E-02  | 4.20E-06 | 2.32E-06 | ND | ND | ND | ND | ND | 1.05E-10 | ND | 0.00E+00 | 2.05E-06 | 9.14E-06  | 1.15E-06  | -8.46E-04 |
| ADP-fossil resources                | MJ                     | 4.66E+03 | 2.38E+01 | 1.39E+02  | 4.82E+03  | 2.38E+01 | 6.84E+00 | ND | ND | ND | ND | ND | 1.69E-04 | ND | 0.00E+00 | 8.87E+00 | 1.08E+01  | 5.52E+00  | -3.57E+02 |
| Water use <sup>5)</sup>             | m <sup>3</sup> e depr. | 1.19E+02 | 1.16E-01 | 5.25E+00  | 1.25E+02  | 1.10E-01 | 1.85E-01 | ND | ND | ND | ND | ND | 1.67E-06 | ND | 0.00E+00 | 4.14E-02 | 1.59E+00  | 7.74E-01  | -5.39E+00 |

1) GWP = Global Warming Potential; 2) EP = Eutrophication potential. Required characterisation method and data are in kg P-eq. Multiply by 3,07 to get PO<sub>4</sub>e; 3) POCP = Photochemical ozone formation; 4) ADP = Abiotic depletion potential; 5) EN 15804+A2 disclaimer for Abiotic depletion and Water use and optional indicators except Particulate matter and Ionizing radiation, human health. The results of these environmental impact indicators shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.

## ADDITIONAL (OPTIONAL) ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2, EF 3.1

| Impact category                  | Unit          | A1       | A2       | A3       | A1-A3    | A4       | A5       | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3       | C4       | D         |
|----------------------------------|---------------|----------|----------|----------|----------|----------|----------|----|----|----|----|----|----------|----|----------|----------|----------|----------|-----------|
| Particulate matter               | Incidence     | 1.55E-05 | 1.50E-07 | 5.19E-07 | 1.62E-05 | 1.49E-07 | 4.54E-08 | ND | ND | ND | ND | ND | 1.44E-13 | ND | 0.00E+00 | 5.11E-08 | 8.45E-08 | 3.22E-08 | -1.83E-06 |
| Ionizing radiation <sup>6)</sup> | kBq<br>I1235e | 2.03E+01 | 2.33E-02 | 1.12E+00 | 2.14E+01 | 1.92E-02 | 2.56E-02 | ND | ND | ND | ND | ND | 5.96E-06 | ND | 0.00E+00 | 7.23E-03 | 1.97E-02 | 6.00E-03 | -4.64E-01 |
| Ecotoxicity (freshwater)         | CTUe          | 7.63E+03 | 3.27E+00 | 7.37E+01 | 7.71E+03 | 4.93E+00 | 8.67E+00 | ND | ND | ND | ND | ND | 1.71E-05 | ND | 0.00E+00 | 1.39E+00 | 5.03E+01 | 3.34E+01 | -2.94E+02 |
| Human toxicity, cancer           | CTUh          | 3.09E-07 | 2.78E-10 | 1.88E-08 | 3.28E-07 | 2.89E-10 | 3.23E-10 | ND | ND | ND | ND | ND | 2.21E-15 | ND | 0.00E+00 | 1.07E-10 | 2.33E-09 | 9.95E-10 | -8.66E-09 |
| Human tox. non-cancer            | CTUh          | 6.41E-06 | 1.51E-08 | 7.48E-08 | 6.50E-06 | 1.41E-08 | 1.75E-08 | ND | ND | ND | ND | ND | 1.05E-13 | ND | 0.00E+00 | 5.57E-09 | 7.46E-08 | 3.92E-08 | -2.02E-07 |
| SQP <sup>7)</sup>                | -             | 9.49E+02 | 2.06E+01 | 1.67E+03 | 2.64E+03 | 2.09E+01 | 5.99E+00 | ND | ND | ND | ND | ND | 1.03E-04 | ND | 0.00E+00 | 5.61E+00 | 5.92E+00 | 3.37E+00 | -6.82E+02 |

6) EN 15804+A2 disclaimer for Ionizing radiation, human health. This impact category deals mainly with the eventual impact of low-dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator; 7) SQP = Land use related impacts/soil quality.

## USE OF NATURAL RESOURCES

| Impact category                    | Unit           | A1       | A2       | A3       | A1-A3    | A4       | A5        | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3        | C4        | D         |
|------------------------------------|----------------|----------|----------|----------|----------|----------|-----------|----|----|----|----|----|----------|----|----------|----------|-----------|-----------|-----------|
| Renew. PER as energy <sup>8)</sup> | MJ             | 3.35E+02 | 3.48E-01 | 7.72E+01 | 4.12E+02 | 3.04E-01 | -2.33E+02 | ND | ND | ND | ND | ND | 5.68E-05 | ND | 0.00E+00 | 1.22E-01 | 5.36E-01  | 1.30E-01  | -9.94E+01 |
| Renew. PER as material             | MJ             | 0.00E+00 | 0.00E+00 | 2.34E+02 | 2.34E+02 | 0.00E+00 | -2.34E+02 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 0.00E+00  | 0.00E+00  | 1.10E+02  |
| Total use of renew. PER            | MJ             | 3.35E+02 | 3.48E-01 | 3.11E+02 | 6.46E+02 | 3.04E-01 | -4.67E+02 | ND | ND | ND | ND | ND | 5.68E-05 | ND | 0.00E+00 | 1.22E-01 | 5.36E-01  | 1.30E-01  | 1.06E+01  |
| Non-re. PER as energy              | MJ             | 3.73E+03 | 2.38E+01 | 1.30E+02 | 3.89E+03 | 2.38E+01 | 6.84E+00  | ND | ND | ND | ND | ND | 1.69E-04 | ND | 0.00E+00 | 8.87E+00 | -3.86E+02 | -3.01E+02 | -3.74E+02 |
| Non-re. PER as material            | MJ             | 6.77E+02 | 0.00E+00 | 2.28E+00 | 6.79E+02 | 0.00E+00 | -1.06E+01 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | -4.01E+02 | -2.68E+02 | 8.67E+01  |
| Total use of non-re. PER           | MJ             | 4.41E+03 | 2.38E+01 | 1.32E+02 | 4.57E+03 | 2.38E+01 | -3.78E+00 | ND | ND | ND | ND | ND | 1.69E-04 | ND | 0.00E+00 | 8.87E+00 | -7.87E+02 | -5.68E+02 | -2.88E+02 |
| Secondary materials                | kg             | 1.91E+00 | 1.04E-02 | 6.74E+00 | 8.66E+00 | 1.02E-02 | 6.49E-03  | ND | ND | ND | ND | ND | 3.02E-08 | ND | 0.00E+00 | 3.97E-03 | 1.31E-02  | 2.80E-03  | 7.07E+00  |
| Renew. secondary fuels             | MJ             | 1.96E-01 | 1.29E-04 | 5.86E+00 | 6.05E+00 | 1.14E-04 | 5.11E-05  | ND | ND | ND | ND | ND | 2.17E-10 | ND | 0.00E+00 | 5.05E-05 | 2.72E-04  | 8.52E-05  | -4.23E-03 |
| Non-ren. secondary fuels           | MJ             | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00  | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 0.00E+00  | 0.00E+00  | 0.00E+00  |
| Use of net fresh water             | m <sup>3</sup> | 2.98E+00 | 3.39E-03 | 1.25E-01 | 3.11E+00 | 3.24E-03 | -1.38E-02 | ND | ND | ND | ND | ND | 3.99E-08 | ND | 0.00E+00 | 1.19E-03 | 2.99E-02  | 5.28E-03  | -1.43E-01 |

8) PER = Primary energy resources.

## END OF LIFE – WASTE

| Impact category     | Unit | A1       | A2       | A3       | A1-A3    | A4       | A5       | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3       | C4       | D         |
|---------------------|------|----------|----------|----------|----------|----------|----------|----|----|----|----|----|----------|----|----------|----------|----------|----------|-----------|
| Hazardous waste     | kg   | 2.81E+01 | 3.84E-02 | 5.57E-01 | 2.87E+01 | 3.88E-02 | 6.41E-02 | ND | ND | ND | ND | ND | 2.60E-07 | ND | 0.00E+00 | 1.54E-02 | 4.22E-01 | 1.99E-01 | -1.39E+00 |
| Non-hazardous waste | kg   | 8.62E+02 | 7.33E-01 | 1.44E+01 | 8.77E+02 | 7.02E-01 | 2.60E+01 | ND | ND | ND | ND | ND | 5.83E-06 | ND | 0.00E+00 | 2.89E-01 | 1.07E+01 | 1.93E+01 | -2.21E+01 |
| Radioactive waste   | kg   | 5.09E-03 | 5.74E-06 | 2.80E-04 | 5.37E-03 | 4.69E-06 | 6.45E-06 | ND | ND | ND | ND | ND | 1.30E-09 | ND | 0.00E+00 | 1.77E-06 | 4.97E-06 | 1.51E-06 | -1.07E-04 |

## END OF LIFE – OUTPUT FLOWS

| Impact category               | Unit | A1       | A2       | A3       | A1-A3    | A4       | A5       | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3       | C4       | D        |
|-------------------------------|------|----------|----------|----------|----------|----------|----------|----|----|----|----|----|----------|----|----------|----------|----------|----------|----------|
| Components for re-use         | kg   | 0.00E+00 | 0.00E+00 | 2.35E-01 | 2.35E-01 | 0.00E+00 | 0.00E+00 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| Materials for recycling       | kg   | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 8.18E+00 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 5.93E+00 | 0.00E+00 | 0.00E+00 |
| Materials for energy rec      | kg   | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| Exported energy               | MJ   | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 1.85E+01 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 1.11E+02 | 0.00E+00 | 0.00E+00 |
| Exported energy – Electricity | MJ   | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 7.78E+00 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 4.68E+01 | 0.00E+00 | 0.00E+00 |
| Exported energy – Heat        | MJ   | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 1.07E+01 | ND | ND | ND | ND | ND | 0.00E+00 | ND | 0.00E+00 | 0.00E+00 | 6.43E+01 | 0.00E+00 | 0.00E+00 |

## ENVIRONMENTAL IMPACTS – EN 15804+A1, CML

| Impact category      | Unit                               | A1       | A2       | A3       | A1-A3    | A4       | A5       | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3       | C4       | D         |
|----------------------|------------------------------------|----------|----------|----------|----------|----------|----------|----|----|----|----|----|----------|----|----------|----------|----------|----------|-----------|
| Global Warming Pot.  | kg CO <sub>2</sub> e               | 3.08E+02 | 1.65E+00 | 9.04E+00 | 3.19E+02 | 1.67E+00 | 1.38E+00 | ND | ND | ND | ND | ND | 6.23E-06 | ND | 0.00E+00 | 6.27E-01 | 2.25E+01 | 1.14E+01 | -2.45E+01 |
| Ozone depletion Pot. | kg CFC <sub>11</sub> e             | 1.01E-05 | 2.15E-08 | 1.98E-07 | 1.04E-05 | 1.97E-08 | 6.32E-09 | ND | ND | ND | ND | ND | 2.59E-13 | ND | 0.00E+00 | 7.08E-09 | 1.38E-08 | 7.28E-09 | -5.61E-08 |
| Acidification        | kg SO <sub>2</sub> e               | 1.48E+00 | 4.94E-03 | 3.06E-02 | 1.51E+00 | 1.00E-02 | 2.07E-03 | ND | ND | ND | ND | ND | 1.84E-08 | ND | 0.00E+00 | 1.61E-03 | 9.71E-03 | 4.49E-03 | -1.12E-01 |
| Eutrophication       | kg PO <sub>4</sub> <sup>3</sup> e  | 1.22E+00 | 1.05E-03 | 1.45E-01 | 1.37E+00 | 1.61E-03 | 1.51E-03 | ND | ND | ND | ND | ND | 3.10E-09 | ND | 0.00E+00 | 3.92E-04 | 3.95E-03 | 2.16E-03 | -1.29E-02 |
| POCP (“smog”)        | kg C <sub>2</sub> H <sub>4</sub> e | 1.08E-01 | 4.06E-04 | 3.82E-03 | 1.12E-01 | 6.43E-04 | 3.57E-04 | ND | ND | ND | ND | ND | 1.27E-09 | ND | 0.00E+00 | 1.44E-04 | 6.54E-04 | 3.22E-04 | -7.47E-03 |
| ADP-elements         | kg Sbe                             | 3.71E-02 | 4.69E-06 | 3.91E-05 | 3.72E-02 | 4.10E-06 | 2.26E-06 | ND | ND | ND | ND | ND | 1.05E-10 | ND | 0.00E+00 | 2.00E-06 | 8.60E-06 | 8.89E-07 | -8.45E-04 |
| ADP-fossil           | MJ                                 | 4.29E+03 | 2.34E+01 | 1.19E+02 | 4.43E+03 | 2.35E+01 | 6.41E+00 | ND | ND | ND | ND | ND | 9.04E-05 | ND | 0.00E+00 | 8.76E+00 | 1.04E+01 | 5.43E+00 | -3.42E+02 |

## ADDITIONAL INDICATOR – GWP-GHG

| Impact category       | Unit                 | A1       | A2       | A3       | A1-A3    | A4       | A5       | B1 | B2 | B3 | B4 | B5 | B6       | B7 | C1       | C2       | C3       | C4       | D         |
|-----------------------|----------------------|----------|----------|----------|----------|----------|----------|----|----|----|----|----|----------|----|----------|----------|----------|----------|-----------|
| GWP-GHG <sup>9)</sup> | kg CO <sub>2</sub> e | 3.10E+02 | 1.66E+00 | 8.97E+00 | 3.21E+02 | 1.68E+00 | 5.59E-01 | ND | ND | ND | ND | ND | 6.25E-06 | ND | 0.00E+00 | 6.31E-01 | 2.25E+01 | 1.14E+01 | -2.47E+01 |

9) This indicator includes all greenhouse gases excluding biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. In addition, the characterisation factors for the flows – CH<sub>4</sub> fossil, CH<sub>4</sub> biogenic and Dinitrogen monoxide – were updated. This indicator is identical to the GWP-total of EN 15804:2012+A2:2019 except that the characterisation factor for biogenic CO<sub>2</sub> is set to zero.



## SCENARIO DOCUMENTATION

### DATA SOURCES

#### Manufacturing energy scenario documentation

1. Heat production, natural gas, at industrial furnace >100kW, Albania, Ecoinvent, 0.0773 kgCO<sub>2</sub>e/MJ
2. Electricity, medium voltage, residual mix, United Kingdom, Ecoinvent, 0.47 kgCO<sub>2</sub>e/kWh

#### Transport scenario documentation - A4 (Transport resources)

1. Market for transport, freight, lorry >32 metric ton, EURO5, 318 km
2. Market for transport, freight, sea, ferry, heavy fuel oil, 54.489 km

#### Transport scenario documentation A4

| Scenario parameter                              | Value    |
|-------------------------------------------------|----------|
| Capacity utilization (including empty return) % | 100      |
| Bulk density of transported products            | 0.00E+00 |
| Volume capacity utilization factor              | 1        |

#### Installation scenario documentation - A5 (Installation waste)

1. Treatment of waste wood, post-consumer, sorting and shredding, Ecoinvent, Materials for recycling, 3.2 kg
2. Treatment of waste wood, untreated, municipal incineration, Ecoinvent, 3.0 kg
3. Exported Energy: Electricity, Ecoinvent, 6.7 MJ
4. Exported Energy: Electricity, Ecoinvent, 1.08 MJ
5. Exported Energy: Thermal, Ecoinvent, 9.2 MJ
6. Exported Energy: Thermal, Ecoinvent, 1.5 MJ
7. Treatment of waste wood, untreated, sanitary landfill, Ecoinvent, 3.8 kg
8. Treatment of waste paper, unsorted, sorting, Ecoinvent, Materials for recycling, 4.98 kg

9. Treatment of waste packaging paper, municipal incineration, Ecoinvent, 0.48 kg
10. Treatment of waste packaging paper, sanitary landfill, Ecoinvent, 0.54 kg

#### Use stages scenario documentation - B6 (Energy data source)

1. Market for electricity, low voltage, United Kingdom, Ecoinvent, 2.4E-5 kWh

#### Use stages scenario documentation - B6-B7 Use of energy and use of water

| Scenario information                                                                                                     | Value                                    |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Ancillary materials specified by material / kg or units as appropriate                                                   | 1                                        |
| Characteristic performance, e.g., energy efficiency, emissions, variation of performance with capacity utilization, etc. | Power consumption is the parameter used. |
| Further assumptions for scenario development, e.g., frequency and period of use, number of occupants                     |                                          |

#### End-of-life scenario documentation - C1-C4 (Data source)

1. Treatment of waste polyethylene, for recycling, unsorted, sorting, Ecoinvent, Materials for recycling, 2.35 kg
2. Treatment of waste plastic, mixture, municipal incineration, Ecoinvent, 4.69 kg
3. Treatment of waste plastic, mixture, municipal incineration, Ecoinvent, 2.35 kg
4. Exported Energy: Thermal, Ecoinvent, 36.714 MJ
5. Exported Energy: Thermal, Ecoinvent, 27.63 MJ
6. Exported Energy: Electricity, Ecoinvent, 26.744 MJ
7. Exported Energy: Electricity, Ecoinvent, 20.088 MJ
8. Treatment of waste plastic, mixture, sanitary landfill, Ecoinvent, 2.35 kg
9. Treatment of waste plastic, mixture, sanitary landfill, Ecoinvent, 0.26 kg
10. Treatment of waste polyurethane, municipal incineration, Ecoinvent, 4.1 kg
11. Treatment of waste polyurethane, municipal incineration, Ecoinvent, 2.05 kg
12. Treatment of waste polyurethane, inert material landfill, Ecoinvent, 2.05 kg
13. Sorting and pressing of iron scrap, Ecoinvent, Materials for recycling, 1.941 kg

- 14. Sorting and pressing of iron scrap, Ecoinvent, Materials for recycling, 0.018 kg
- 15. Treatment of scrap steel, inert material landfill, Ecoinvent, 0.49 kg
- 16. Treatment of scrap steel, inert material landfill, Ecoinvent, 0.012 kg
- 17. Sorting and pressing of iron scrap, Ecoinvent, Materials for recycling, 1.62 kg
- 18. Treatment of scrap steel, municipal incineration, Ecoinvent, 0.54 kg
- 19. Treatment of scrap steel, inert material landfill, Ecoinvent, 0.54 kg

| Scenario information                     | Value                                                                                                                                                                                                                                                        |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scenario assumptions e.g. transportation | ABS: 20%Recycled, 60%<br>Incinerated. PU: 75%<br>incinerated, 25% landfill.<br>Steel: 80% Recycled 20%<br>Landfill Other Non-Ferrous<br>Metals: 60% Recycled, 20%<br>incinerated, 20% landfill. PCB:<br>100% Landfill. Copper: 60%<br>Recycled, 40% landfill |

## THIRD-PARTY VERIFICATION STATEMENT

EPD Hub declares that this EPD is verified in accordance with ISO 14025 by an independent, third-party verifier. The project report on the Life Cycle Assessment and the report(s) on features of environmental relevance are filed at EPD Hub. EPD Hub PCR and ECO Platform verification checklist are used.

EPD Hub is not able to identify any unjustified deviations from the PCR and EN 15804+A2 in the Environmental Product Declaration and its project report.

EPD Hub maintains its independence as a third-party body; it was not involved in the execution of the LCA or in the development of the declaration and has no conflicts of interest regarding this verification.

The company-specific data and upstream and downstream data have been examined as regards plausibility and consistency. The publisher is responsible for ensuring the factual integrity and legal compliance of this declaration.

The software used in creation of this LCA and EPD is verified by EPD Hub to conform to the procedural and methodological requirements outlined in ISO 14025:2010, ISO 14040/14044, EN 15804+A2, and EPD Hub Core Product Category Rules and General Program Instructions.

### Verified tools

Tool verifier: Magaly Gonzalez Vazquez

Tool verification validity: 27 March 2025 - 26 March 2028

Yazan Badour as an authorized verifier for EPD Hub Limited 13.03.2026



## ANNEX 1 PART NUMBERS

Annex I: List of all parts covered by this EPD

| Part Number | Part Description                                               | GTIN          | Weight |
|-------------|----------------------------------------------------------------|---------------|--------|
| 443319P     | Lo-Carbon Sentinel Kinetic BH with 2.5m Fly Lead               | 5015135983840 | 15     |
| 443319LP    | Lo-Carbon Sentinel Kinetic BH with 2.5m Fly Lead (Left Handed) | 5015135983857 | 15     |
| 438222L     | Lo-Carbon Sentinel Kinetic B Left Hand                         | 5015135062170 | 15     |
| 438222      | Lo-Carbon Sentinel Kinetic B                                   | 5015135382223 | 15     |
| 443319L     | Lo-Carbon Sentinel Kinetic BH Left                             | 5015135062187 | 15     |
| 443319      | Lo-Carbon Sentinel Kinetic BH                                  | 5015135433192 | 15     |
| 1004000002  | KIN-B-R / Kinetic B met bypass                                 | 5015135010485 | 15     |
| 1004000006  | KIN-BH-R / Kinetic B met bypass en hygrostaat                  | 5015135010492 | 15     |
| 1004000008  | KIN-B-L / Kinetic B met bypass                                 | 5015135010508 | 15     |
| 1004000009  | KIN-BH-L / Kinetic B met bypass en hygrostaat                  | 5015135010430 | 15     |
| 8000000046  | KIN-B-R / Kinetic B met bypass                                 | 5015135994341 | 15     |
| 8000000286  | KIN-B-L / Kinetic B met bypass                                 | 5015135994334 | 15     |
| 8000000287  | KIN-BH-R / Kinetic B met bypass en hygrostaat                  | 5015135994327 | 15     |
| 8000000288  | KIN-BH-L / Kinetic B met bypass en hygrostaat                  | 5015135994310 | 15     |
| E150019     | MVHR 75BH HEAT RECOVERY UNIT. (KINETIC BH)                     | 5015135010515 | 15     |
| E150076     | VENT-AXIA MVHR UNIT RH                                         | 5015135010454 | 15     |
| E150077     | VENT-AXIA MVHR UNIT LH                                         | 5015135010461 | 15     |
| 40-8610-L   | KINETIC PLUS L/H ENTHALPY CELL FOR ALLVOTECH (BH)              | 5015135010553 | 15     |
| 40-8610-R   | KINETIC PLUS R/H ENTHALPY CELL FOR ALLVOTECH (BH)              | 5015135010478 | 15     |
| E150020     | MVHR 135BH HEAT RECOVERY UNIT (KINETIC PLUS B)                 | 5015135010560 | 15     |
| 443028P     | Sentinel Kinetic Plus B (RH) - 2.5m fly lead                   | 5015135983888 | 15     |
| 443028LP    | Sentinel Kinetic Plus B (LH) - 2.5m fly lead                   | 5015135983895 | 15     |
| 8000000057  | KIN PLUS-BH-L / Kinetic PLUS bypass + hygrostaat               | 5015135994297 | 15     |
| 1004000013  | KIN PLUS-BH-L / Kinetic PLUS bypass + hygrostaat               | 5015135010577 | 15     |

| Part Number | Part Description                                    | GTIN          | Weight |
|-------------|-----------------------------------------------------|---------------|--------|
| 8000000056  | KIN PLUS-BH-R / Kinetic PLUS bypass + hygrostaat    | 5015135994303 | 15     |
| 1004000011  | KIN PLUS-BH-R / Kinetic PLUS bypass + hygrostaat    | 5015135010584 | 15     |
| 408169P     | Sentinel Kinetic F (LH) - 2.5m fly lead             | 5015135983871 | 19     |
| 408167      | Lo-Carbon Sentinel Kinetic FH                       | 5015135081676 | 19     |
| 8640-L      | KINETIC FHL L/H WITH ENTHALPY ALLVOTECH             | 5015135010522 | 19     |
| 8640-R      | KINETIC FH R/H WITH ENTHALPY ALLVOTECH              | 5015135010539 | 19     |
| 408167P     | Sentinel Kinetic F (RH) - 2.5m fly lead             | 5015135983864 | 19     |
| 8000000053  | KIN-FH-L / Kinetic F met vochtsensor                | 5015135994358 | 19     |
| 1004000123  | KIN-FH-L / Kinetic F met vochtsensor                | 5015135010447 | 19     |
| 8000000052  | KIN-FH-R / Kinetic F met vochtsensor                | 5015135994365 | 19     |
| 1004000122  | KIN-FH-R / Kinetic F met vochtsensor                | 5015135010546 | 19     |
| 443028L     | Lo-Carbon Sentinel Kinetic Plus Left B              | 5015135062194 | 24     |
| 443028      | Lo-Carbon Sentinel Kinetic Plus B                   | 5015135430283 | 24     |
| 443935L     | SENTINEL KINETIC PLUS (D) LEFT HANDED VERSION       | 5015135068608 | 24     |
| 443935      | SENTINEL KINETIC PLUS (D) DUTCH/FRENCH LEAFLETS     | 5015135439354 | 24     |
| 443938      | SENTINEL KINETIC PLUS (C) CZECH VERSION             | 5015135439385 | 24     |
| 408169      | Lo-Carbon Sentinel Kinetic FH (Left Handed)         | 5015135081690 | 26     |
| 417798      | Sentinel Kinetic FH - Danish Version (Left Handed)  | 5015135010614 | 26     |
| 417799      | Sentinel Kinetic FH – Danish Version (Right Handed) | 5015135010621 | 26     |
| 408451      | Sentinel Kinetic High Flow Left                     | 5015135084516 | 38     |
| 408449      | Sentinel Kinetic High Flow Right                    | 5015135084493 | 38     |
| 408449P     | Sentinel Kinetic High Flow (RH) - 2.5m fly lead     | 5015135983901 | 38     |
| 476781      | HIGH FLOW RH ENTHALPY SENTINEL KINETIC              | 5015135767815 | 38     |
| 476782      | HIGH FLOW LH ENTHALPY SENTINEL KINETIC              | 5015135767822 | 38     |
| 408451P     | Sentinel Kinetic High Flow (LH) - 2.5m fly lead     | 5015135983918 | 38     |

## ANNEX 2 SCALING TABLE

Annex II – Below is a list of ventilation products, their corresponding mass and the GWP-total per item for production stages A1-A3.

| Part Number | Mass (Kg) | A1-A3       |             |           |
|-------------|-----------|-------------|-------------|-----------|
|             |           | EN 15804+A1 | EN 15804+A2 |           |
|             |           | GWP         | GWP total   | GWP total |
| Reference   | 1         | 11.229      | 0.343       | 0.343     |
| 443319P     | 15        | 168.439     | 5.138       | 5.138     |
| 443319LP    | 15        | 168.439     | 5.138       | 5.138     |
| 438222L     | 15        | 168.439     | 5.138       | 5.138     |
| 438222      | 15        | 168.439     | 5.138       | 5.138     |
| 443319L     | 15        | 168.439     | 5.138       | 5.138     |
| 443319      | 15        | 168.439     | 5.138       | 5.138     |
| 1004000002  | 15        | 168.439     | 5.138       | 5.138     |
| 1004000006  | 15        | 168.439     | 5.138       | 5.138     |
| 1004000008  | 15        | 168.439     | 5.138       | 5.138     |
| 1004000009  | 15        | 168.439     | 5.138       | 5.138     |
| 8000000046  | 15        | 168.439     | 5.138       | 5.138     |
| 8000000286  | 15        | 168.439     | 5.138       | 5.138     |
| 8000000287  | 15        | 168.439     | 5.138       | 5.138     |
| 8000000288  | 15        | 168.439     | 5.138       | 5.138     |
| E150019     | 15        | 168.439     | 5.138       | 5.138     |
| E150076     | 15        | 168.439     | 5.138       | 5.138     |
| E150077     | 15        | 168.439     | 5.138       | 5.138     |
| 40-8610-L   | 15        | 168.439     | 5.138       | 5.138     |
| 40-8610-R   | 15        | 168.439     | 5.138       | 5.138     |
| E150020     | 15        | 168.439     | 5.138       | 5.138     |

| Part Number | Mass (Kg) | A1-A3       |             |           |
|-------------|-----------|-------------|-------------|-----------|
|             |           | EN 15804+A1 | EN 15804+A2 |           |
|             |           | GWP         | GWP total   | GWP total |
| 443028P     | 15        | 168.439     | 5.138       | 5.138     |
| 443028LP    | 15        | 168.439     | 5.138       | 5.138     |
| 8000000057  | 15        | 168.439     | 5.138       | 5.138     |
| 1004000013  | 15        | 168.439     | 5.138       | 5.138     |
| 8000000056  | 15        | 168.439     | 5.138       | 5.138     |
| 1004000011  | 15        | 168.439     | 5.138       | 5.138     |
| 408169P     | 19        | 213.356     | 6.508       | 6.508     |
| 408167      | 19        | 213.356     | 6.508       | 6.508     |
| 8640-L      | 19        | 213.356     | 6.508       | 6.508     |
| 8640-R      | 19        | 213.356     | 6.508       | 6.508     |
| 408167P     | 19        | 213.356     | 6.508       | 6.508     |
| 8000000053  | 19        | 213.356     | 6.508       | 6.508     |
| 1004000123  | 19        | 213.356     | 6.508       | 6.508     |
| 8000000052  | 19        | 213.356     | 6.508       | 6.508     |
| 1004000122  | 19        | 213.356     | 6.508       | 6.508     |
| 443028L     | 24        | 269.503     | 8.221       | 8.221     |
| 443028      | 24        | 269.503     | 8.221       | 8.221     |
| 443935L     | 24        | 269.503     | 8.221       | 8.221     |
| 443935      | 24        | 269.503     | 8.221       | 8.221     |
| 443938      | 24        | 269.503     | 8.221       | 8.221     |
| 408169      | 26        | 291.961     | 8.906       | 8.906     |
| 417798      | 26        | 291.961     | 8.906       | 8.906     |
| 417799      | 26        | 291.961     | 8.906       | 8.906     |
| 408451      | 38        | 426.713     | 13.017      | 13.017    |

| Part Number | Mass (Kg) | A1-A3       |             |           |
|-------------|-----------|-------------|-------------|-----------|
|             |           | EN 15804+A1 | EN 15804+A2 |           |
|             |           | GWP         | GWP total   | GWP total |
| 408449      | 38        | 426.713     | 13.017      | 13.017    |
| 408449P     | 38        | 426.713     | 13.017      | 13.017    |
| 476781      | 38        | 426.713     | 13.017      | 13.017    |
| 476782      | 38        | 426.713     | 13.017      | 13.017    |
| 408451P     | 38        | 426.713     | 13.017      | 13.017    |